Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/14/2007
PI Name:	Golden, Barbara L. Ph.D.		
Project Title:	Engineering a Ribozyme for Diffraction Properties		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-03
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG8-1833		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	In vitro evolution will be used to develop ribozymes that are thermostable. The thermostable ribozymes will be crystallized. X-ray diffraction analysis will be used to characterize the diffraction characteristics of the crystals. We hope that this study will result in a detailed view of a ribozyme active site at atomic resolution. Furthermore, we hope to develop a generalized strategy for examining the atomic-resolution of multidomain ribosomes. To develop methods in improving X-ray diffraction from RNA crystals. We are trying to understand the structure and function of self-splicing group I introns at the atomic level using X-ray crystallography. These studies are hampered by the low diffraction limits that are typical of crystals formed by multi-domain RNAs. We wish to develop a general method for 'designing' RNAs that are more suited for analysis by X-ray crystallography, allowing a more varied structural database of RNA active sites to be obtained in our lab and in		

	other laboratories.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research involves studying how RNA molecules can fold into structures capable of biological catalysis. These studies provide insight into biological catalysts that can be harnessed as tools in our molecular biology toolkit. Life on earth today may have evolved from an RNA world, in which RNA encoded genetic material and RNA catalysts performed the chemical reaction necessary for self-replication. Study of these molecule may provide a glimpse into the development of early self-replicating systems on Earth.
Task Progress:	Introduction RNA molecules can perform a great variety of biological functions within the cell. RNAs have the ability to encode genetic information (like DNA), to achieve molecular recognition and perform biological catalysis (like proteins). These abilities of RNA to function both as genetic material and as catalysts has led to the speculation that RNA, or chemically related molecules, were the first self-replicating molecules on Earth. In contrast to the wealth of structural information currently available from the structural genomics projects, at the time the grant was awarded, there was little structural data available for catalytic RNAs. We are particularly interested in studying large, mulitdomain RNA catalysts that, like most protein catalysts, have actives sites large enough to recognize substrates in trans and to orchestrate complex reactions. The molecule we were studying is the self-splicing group I intron from Tetrahymena thermophila. Group I introns are RNA molecules catalyze two sequential phosphotransesterifcation reactions that result in excision of the intron and ligation of the surrounding RNA exons. Key steps in this chemical reaction are recognition of the 5'-splice site and binding of guanosine substrate and activation of this substrate for attack on the phosphate that defines the 5'-splice site. At the start of this project, we were studying an intron from Tetrahymena thermophila and we generated crystals of a molecule that spanned two of the three conserved domains of this group I intron (TetP3-P9 ribozyme). These crystals diffracted X-rays to 5 Å and a low resolution structure of this molecule was determined. Standard crystallographic manipulations did not result in an improvement of the diffraction properties of these crystals and we were unable to analyze this structure at a resolution sufficient for an in depth understanding of the structural basis for group I intron catalysis. The goals of this project were to explore methods for improving the diffraction qualities of these RNA crystals by performing the following Specific Aims: (1) Introduction of mutations known to enhance activity at low divalent ion concentrations, (2) In vitro evolution to stabilize the ribozyme core and (3) Microgravity to improve crystal quality. Progress on Specific Aim 1. Introduction of mutations known to enhance activity at low divalent ion concentrations Joyce and colleagues had used in vitro evolution to identify sequence variants that could function in the presence of calcium ion. Group I introns are metalloenzymes that require the presence of divalent metal ions (usually magnesium). These metal ions are needed to stabilize the three dimensional structure of compactly folded RNAs, and, in the case of group I introns, participate directly in catalysis. Calcium ions facilitate RNA folding, but cannot support catalysis in the wild-type enzyme. The result of this in vitro selection protocol were group I introns that could function in lower concentrations of magnesium than wild-type RNAs. This property suggested that the artificial variants were actually more stable that the wild-type sequence. Since thermostability often improves the crystallization properties and diffraction properties of protein molecules, we hypothesized that these mutations might stabilize the three dimensional structure of the Tetrahymena inton and thereby improve the diffraction properties of the crystals. The Joyce mutations were introduced into the TetP3-P9 ribozyme and these variants were crystallized. No improvement in the crystallization or diffraction properties were obtained. Progress on Specific Aim 2. In vitro evolution to stabilize the ribozyme core. Protein crystallographers have found that thermostable variants of proteins often crystallize more readily, and the crystals obtained often diffract to higher resolution than their mesophilic counterparts. In this aim, we sought to identify thermostable group I introns that would have improved diffraction properties relative to the wild-type sequence. Unfortunately, none of the known thermophilic organisms possess group I intron sequences within their genomes. Thus, we proposed to use artificial evolution to identify sequence variants of the Tetrahymena ribozyme that would increase the stability of the tertiary structure of this RNA. We ultimately used an alternate strategy to achieve the goals of this proposal. We identified a group I intron from bacteriophage Twort that had the properties necessary for structural analysis. First, the Twort ribozyme is thermostable, retaining its ability to self-splice at temperatures has high as 55oC. Second, this RNA produces crystals that diffract X-rays to a maximal resolution of 3.6 Å, a significant improvement from the 5 Å diffraction produced by the TetP3-P9 ribozyme crystals. Third, the Twort ribozyme spanned the entire intron, including the 5' splice site, while the TetP3-P9 ribozyme crystals spanned only two of the three conserved domains. Finally, crystallization of the Twort ribozyme is rapid and efficient, facilitating achievement of Specific Aim 3. The structure of the Twort ribozyme was solved using X-ray crystallography. Details are provided in the attached manuscripts. Progress on Specific Aim 3. Microgravity to improve crystal quality. We pursued microgravity crystallization experiments in an effort to improve the diffraction properties of the Twort ribozyme crystals. Crystallization experiments were flown on missions UF-2 and 9A and were performed with the expertise of Lisa Monaco. In initial crystallization trials, we learned that the PCAM apparatus used to hold and crystallize samples during flights were not completely sealed when MPD was used as a precipitant. Measurable sample loss occurred prior to activation of the PCAM, resulting in premature crystallization of the sample. We developed a new protocol for crystallization of the Twort ribozyme using ammonium sulfate as precipitant, providing better stability in the PCAM apparatus, and improving the efficiency of crystallization. Unfortunately, about this time, we lost the space shuttle Discovery and microgravity experiments were halted. Working with Lisa Monaco, we pursued "lab on a chip" crystallization experiments and her laboratory was able to successfully crystallize the Twort ribozyme using this strategy. This methodology may ultimately prove a superior method for microgravity crystallization experiments.

Bibliography Type:	Description: (Last Updated: 10/21/2009)
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Articles in Peer-reviewed Journals	Chase E, Golden BL. "Crystallization and preliminary diffraction analysis of a group I ribozyme from bacteriophage Twort. " Acta Crystallograph Sect F Struct Biol Cryst Commun. 2005 Jan 1;61(Pt 1):71-4. PMID: 16508095 , Jan-2005
Articles in Peer-reviewed Journals	Golden BL, Kim H, Chase E. "Crystal structure of a phage Twort group I ribozyme-product complex. " Nat Struct Mol Biol. 2005 Jan;12(1):82-9. Epub 2004 Dec 5. PMID: 15580277 , Jan-2005
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Books/Book Chapters	Golden BL. "Ribozyme Structural Elements: Group I Introns." in "Encyclopedia of Biological Chemistry " Ed. W.J. Lennarz & M.D. Lane. Amsterdam ; Boston : Elsevier, 2004. Vol. 4, pp 738-742., Jan-2004